

NAME: _____

PERIOD: _____

Counts or Conditional Percents?

AP Statistics · Unit 2 · Topic 2.1

[STAR] THE PROBLEM

Suppose a counselor records whether each of 200 students plays a school sport and whether the student usually gets at least 8 hours of sleep on a school night.

Kai: *Seventy-five nonathletes but only 35 athletes get at least 8 hours, so playing a sport goes with less sleep.*

Rosa: *Those counts compare groups of different sizes; 35 out of 50 may tell a different story from 75 out of 150.*

Usual sleep on a school night

Sport?	≥ 8 h	< 8 h	Total
Plays	35	15	50
Does not	75	75	150
Total	110	90	200

FIRST TAKE — one sentence before you work the parts

Which student's reasoning seems more useful for deciding whether sport participation and sleep are related? Name one table detail.

[BOOK] COMPARE WITHIN GROUPS (keep on the page)

A **two-way table** records combinations of two categorical variables. A raw count answers *how many*; it does not by itself answer *how likely within a group*. To compare groups of different sizes, compute a **conditional relative frequency**: cell count $\div$ total for that row or column. If the conditional distributions differ, the variables appear associated. Association from observed data does not by itself establish cause.

Work (a)–(c).

DOK 1 (a) Name the two categorical variables.

DOK 2 (b) Find the percent who get at least 8 hours of sleep in each sport group. Show the two fractions.

DOK 3 (c) Whose reasoning is better supported? Use both percentages and explain why comparing 75 with 35 can mislead. State whether this table shows that playing a sport causes more sleep. Write 3–4 sentences.

Frame: The variables appear _____ because _____% of athletes versus _____% of nonathletes _____.

Frame: The 75-to-35 comparison misleads because _____, and the table cannot establish cause because _____.

Turn this sheet in whenever you finish — bonus credit. Part (c) is scored E / P / I.